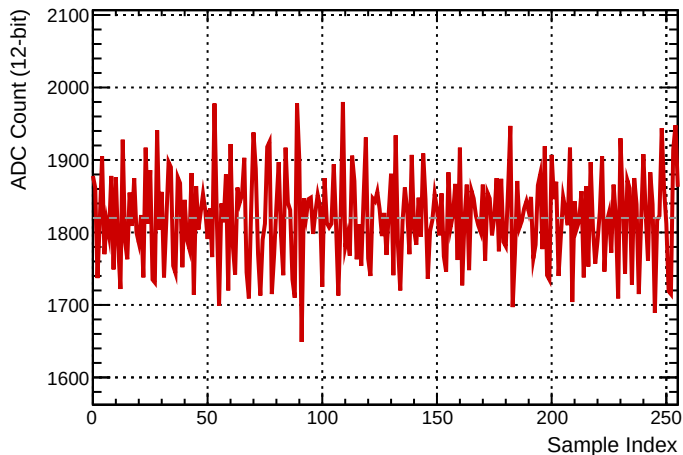
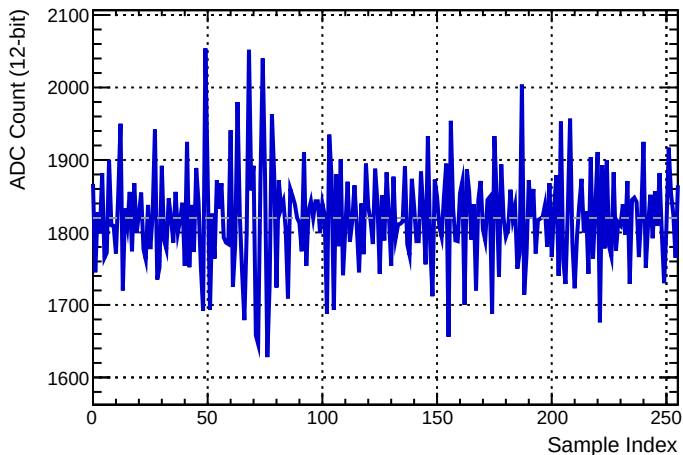


AI Trigger System -- Simulation Results

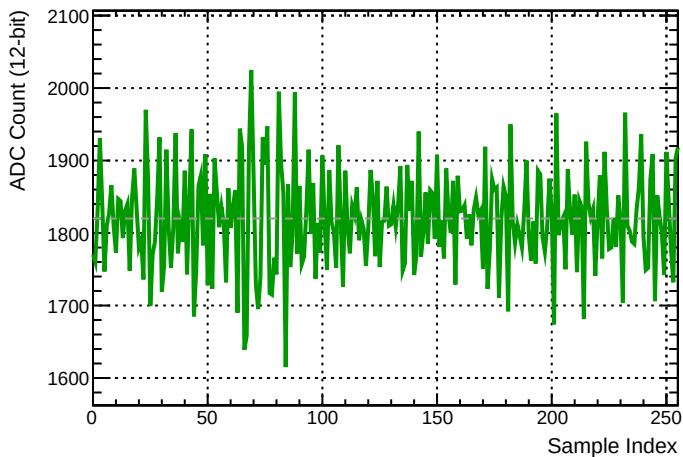
20 injected events | 2 triggers | 20 CNN outputs

62.5 MHz fast clock | 200 MHz CNN clock | 12-bit ADC

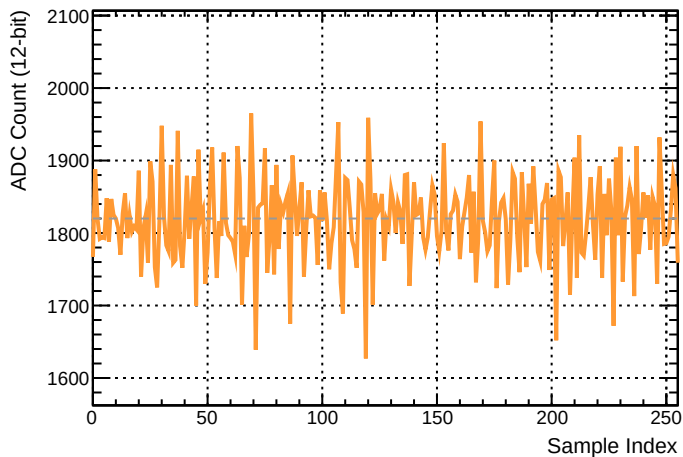
Ch 0 | Event #9 | Label: **SIGNAL** | CNN Score: 5212

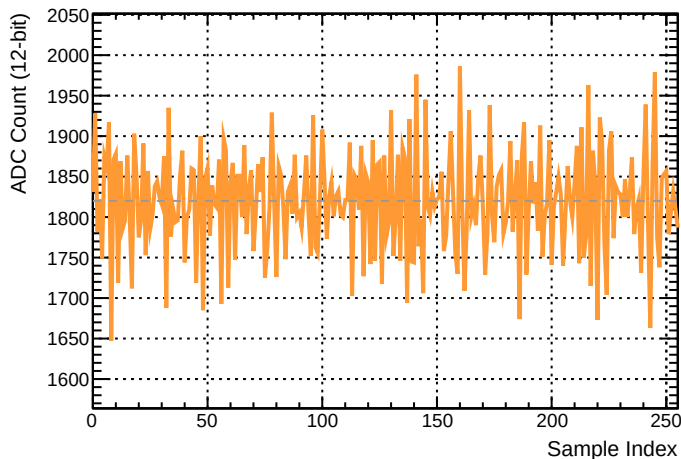
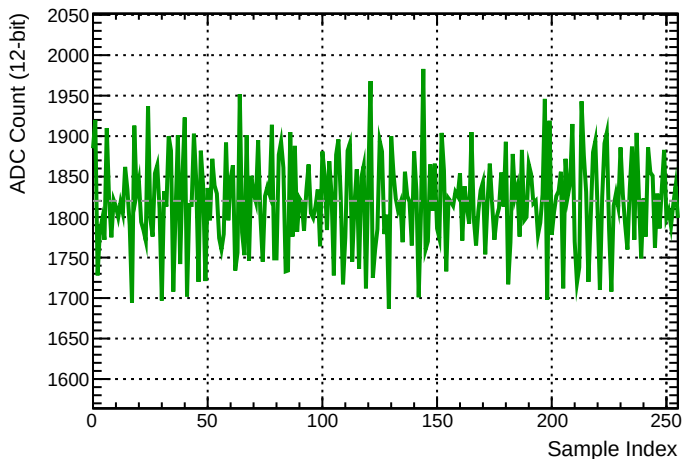
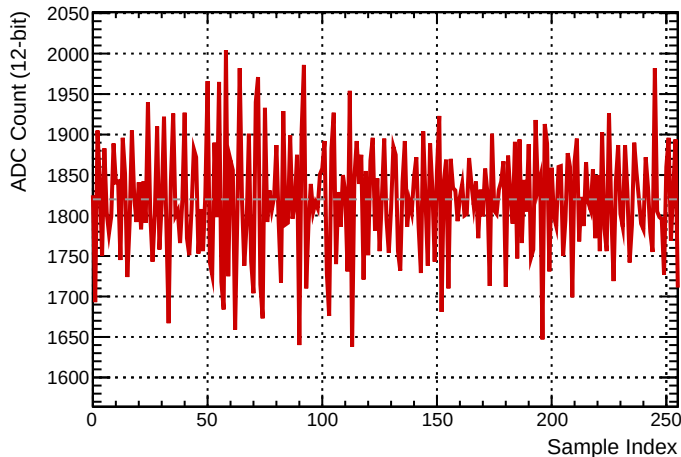
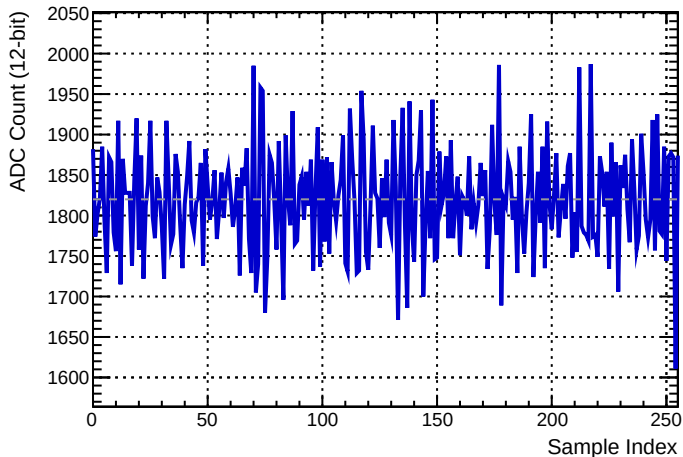


Ch 2 | Event #9 | Label: **SIGNAL** | CNN Score: 5212

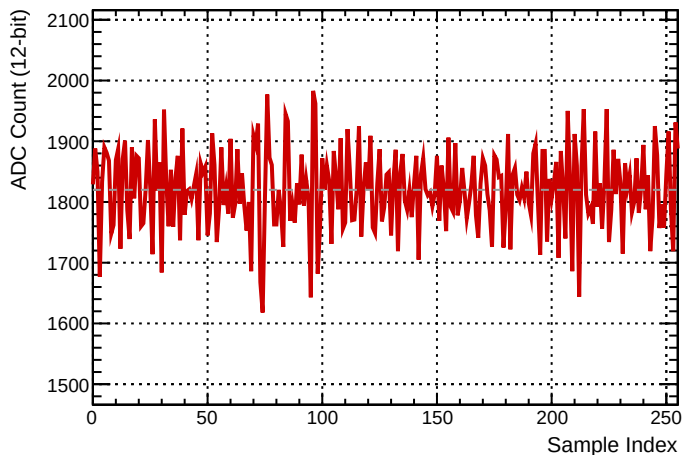
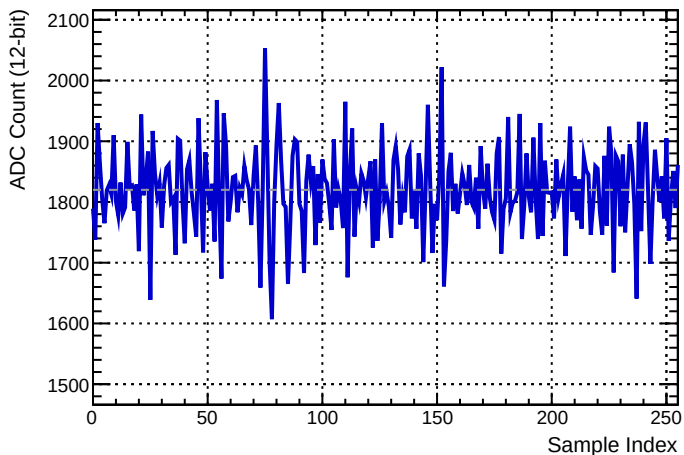


Ch 3 | Event #9 | Label: **SIGNAL** | CNN Score: 5212

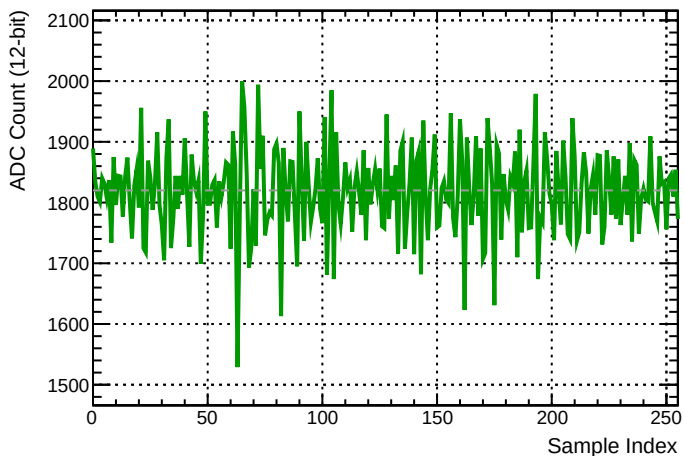




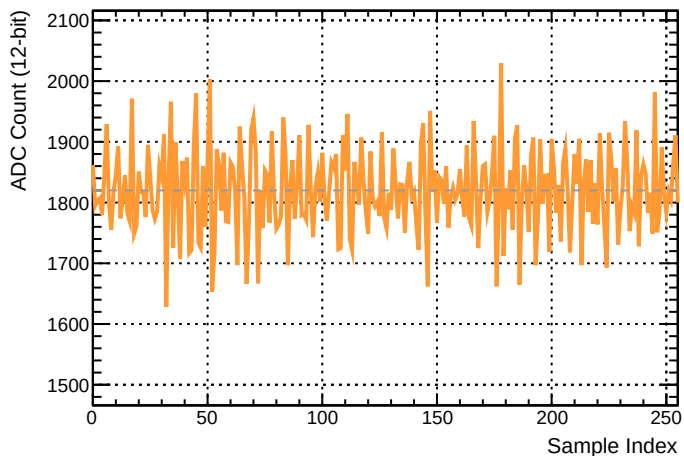
Ch 0 | Event #17 | Label: **SIGNAL** | CNN Score: 5315



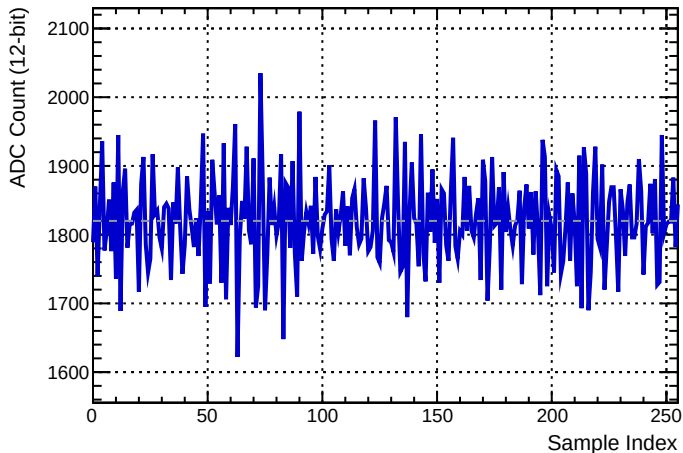
Ch 2 | Event #17 | Label: **SIGNAL** | CNN Score: 5315



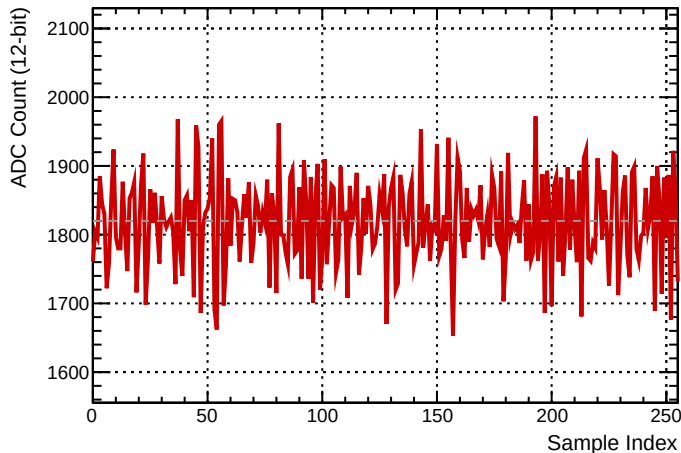
Ch 3 | Event #17 | Label: **SIGNAL** | CNN Score: 5315



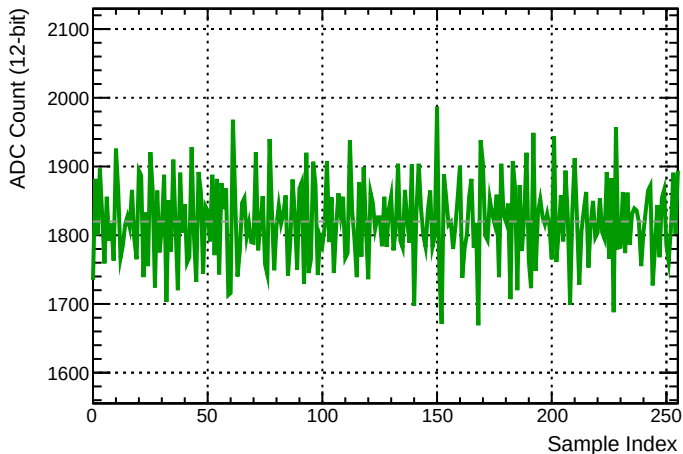
Ch 0 | Event #964 | Label: Background | Sim Test Set | Dataset: 5000 | Event #964 | Label: Background | CNN Score: 5324



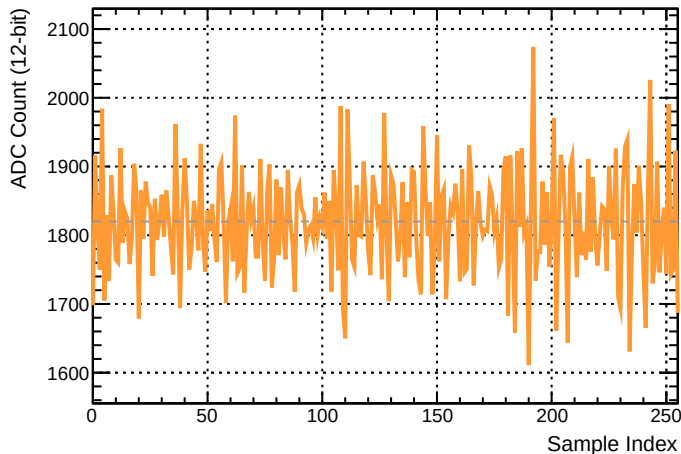
Ch 1 | Event #964 | Label: Background | Sim Test Set | Dataset: 5000 | Event #964 | Label: Background | CNN Score: 5324



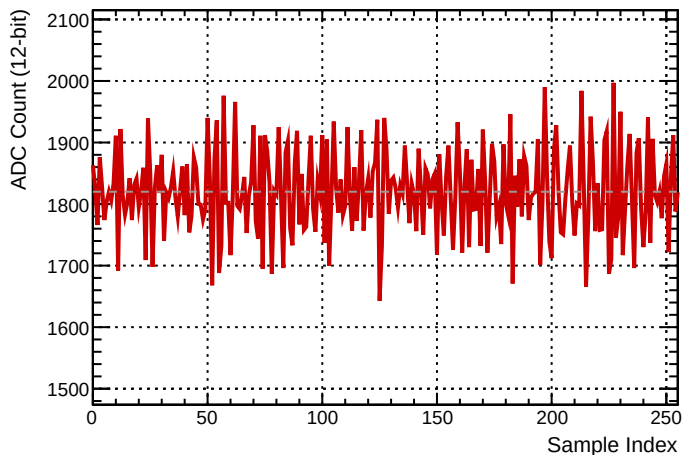
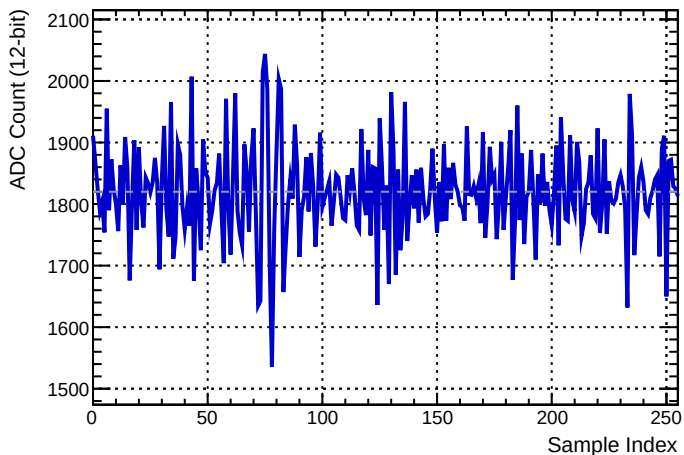
Ch 2 | Event #964 | Label: Background | Sim Test Set | Dataset: 5000 | Event #964 | Label: Background | CNN Score: 5324



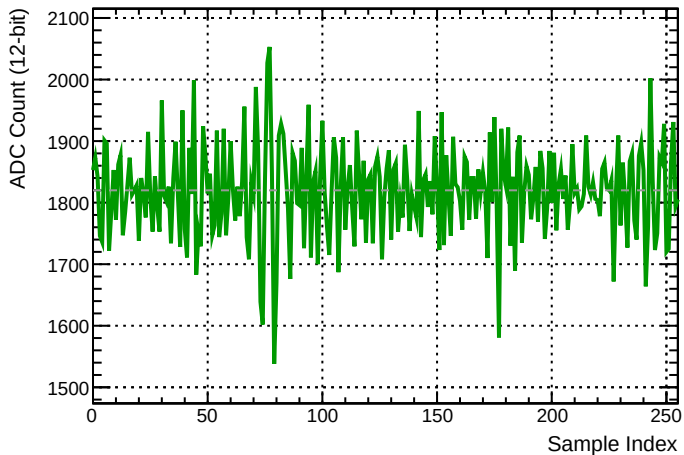
Ch 3 | Event #964 | Label: Background | Sim Test Set | Dataset: 5000 | Event #964 | Label: Background | CNN Score: 5324



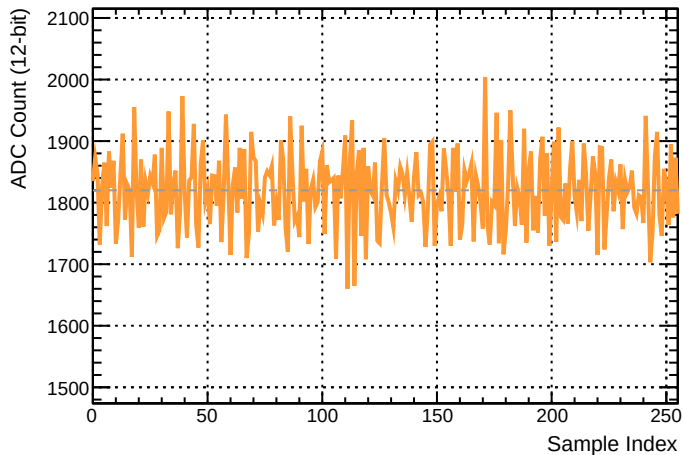
Ch 0 | Event #38 | Label: **SIGNAL** | CNN Score: 5680



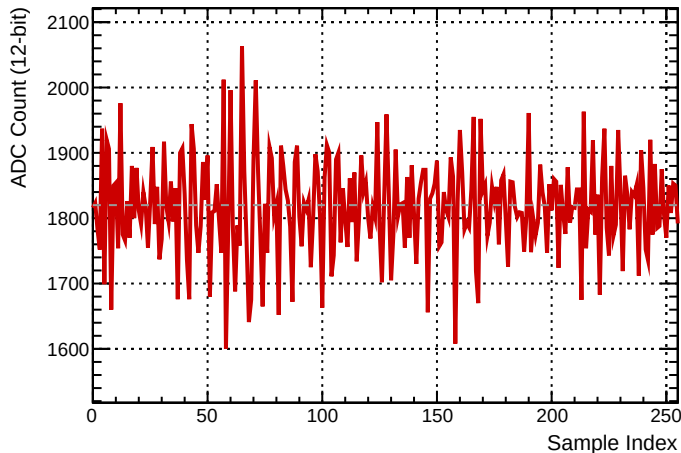
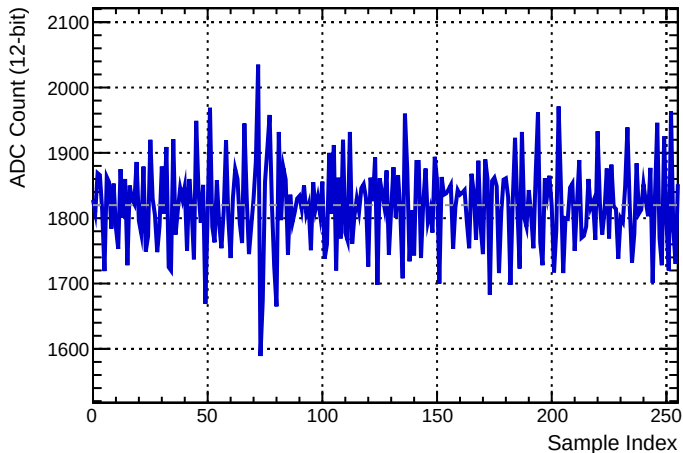
Ch 2 | Event #38 | Label: **SIGNAL** | CNN Score: 5680



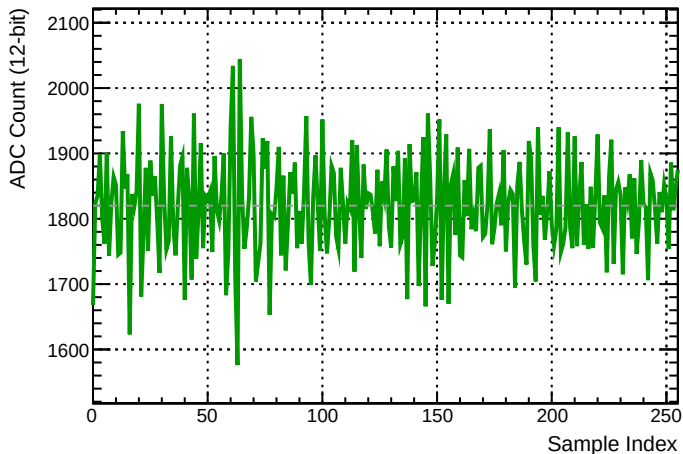
Ch 3 | Event #38 | Label: **SIGNAL** | CNN Score: 5680



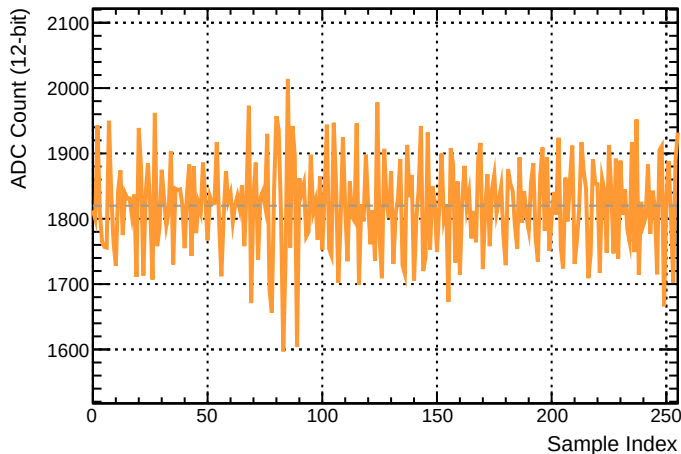
Ch 0 | Event #881 | Label: **SIGNAL** | CNN Score: 4869



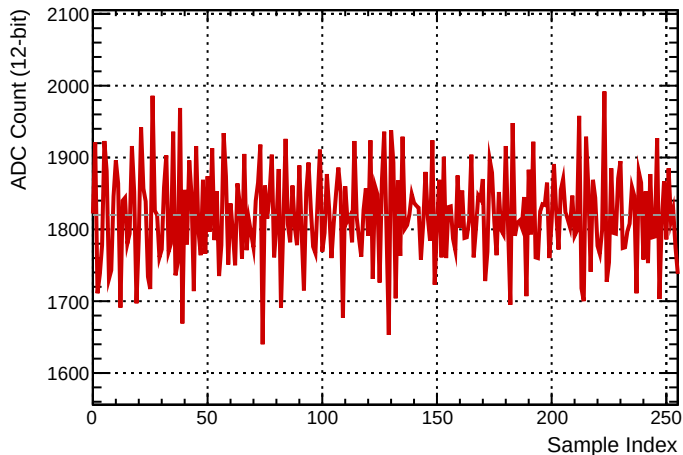
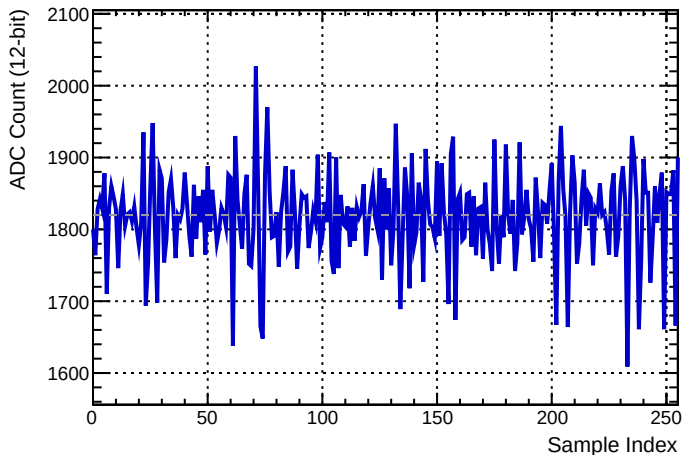
Ch 2 | Event #881 | Label: **SIGNAL** | CNN Score: 4869



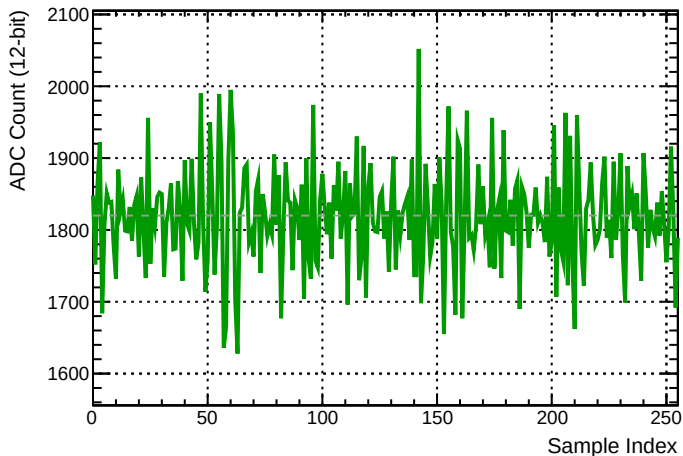
Ch 3 | Event #881 | Label: **SIGNAL** | CNN Score: 4869



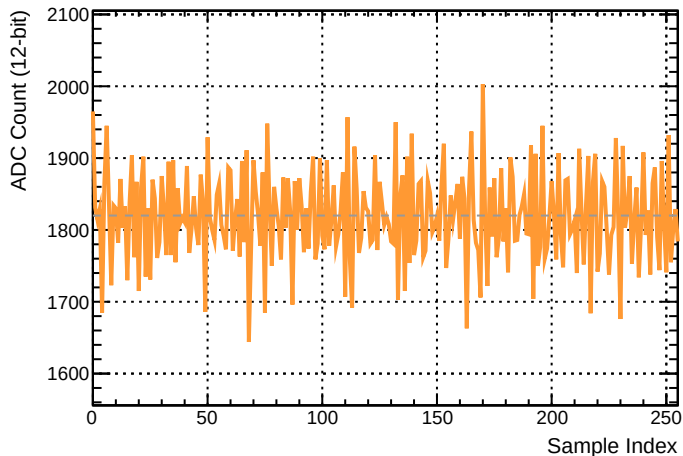
Ch 0 | Event #39 | Label: **SIGNAL** | CNN Score: 5227



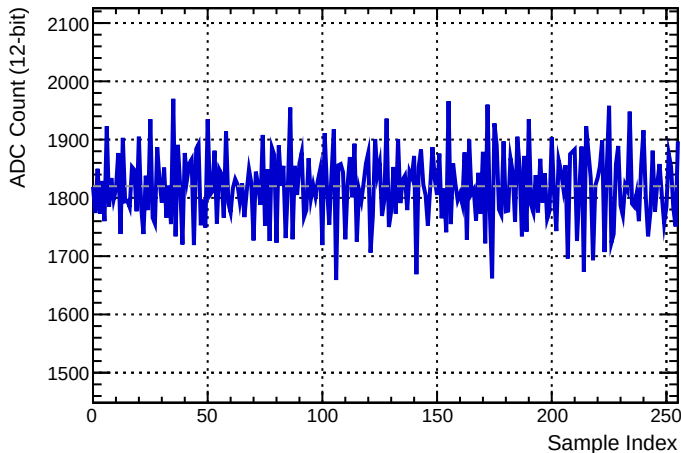
Ch 2 | Event #39 | Label: **SIGNAL** | CNN Score: 5227



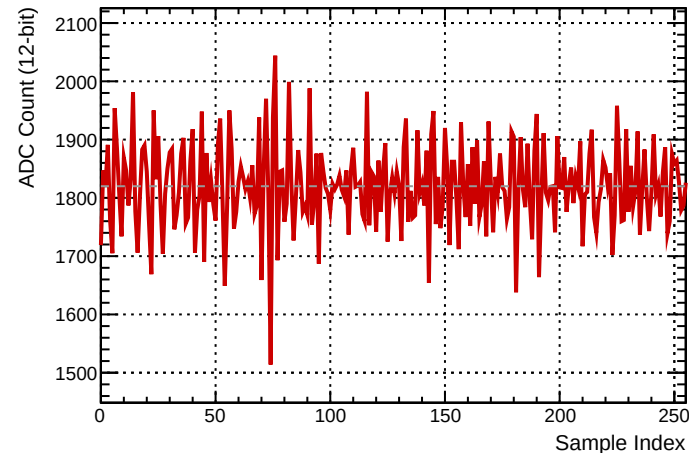
Ch 3 | Event #39 | Label: **SIGNAL** | CNN Score: 5227



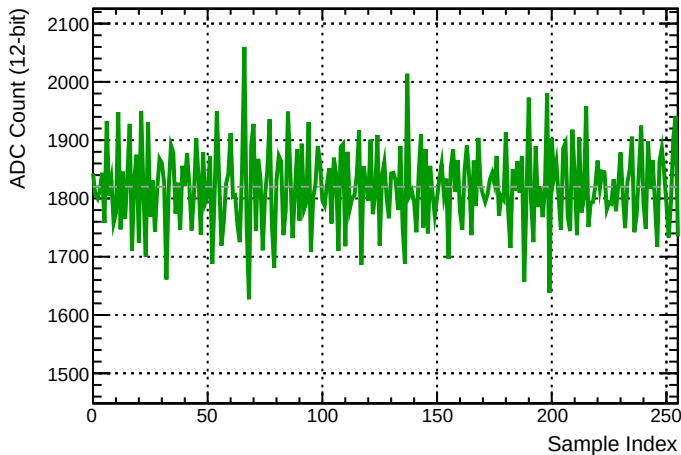
Ch 0 | Event #893 | Label: Background | CNN Score: 4823



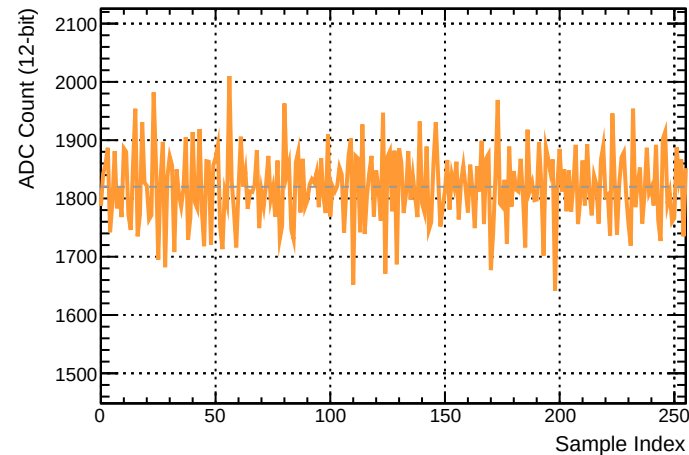
Ch 1 | Event #893 | Label: Background | CNN Score: 4823



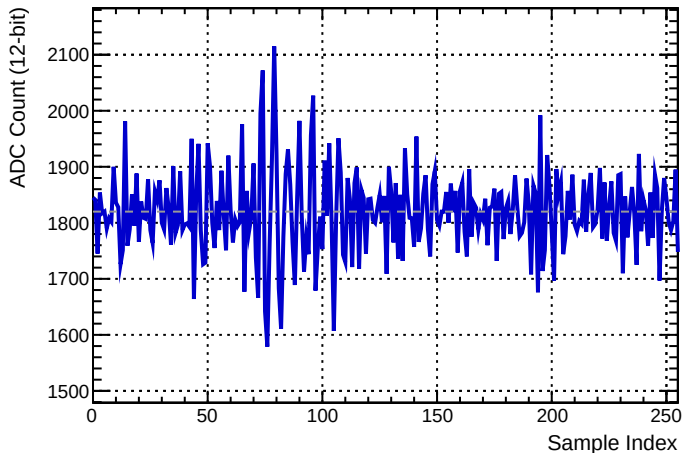
Ch 2 | Event #893 | Label: Background | CNN Score: 4823



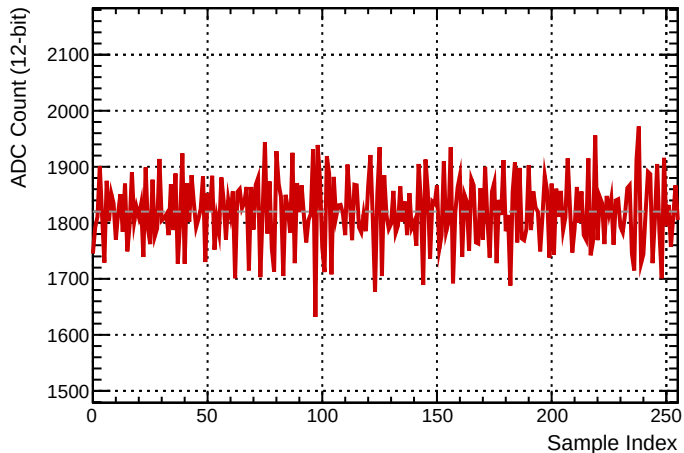
Ch 3 | Event #893 | Label: Background | CNN Score: 4823



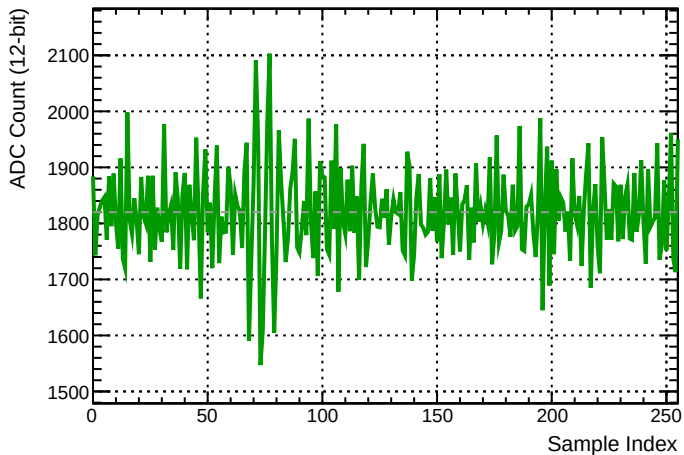
Ch 0 | Event #40 | Label: **SIGNAL** | CNN Score: 5474



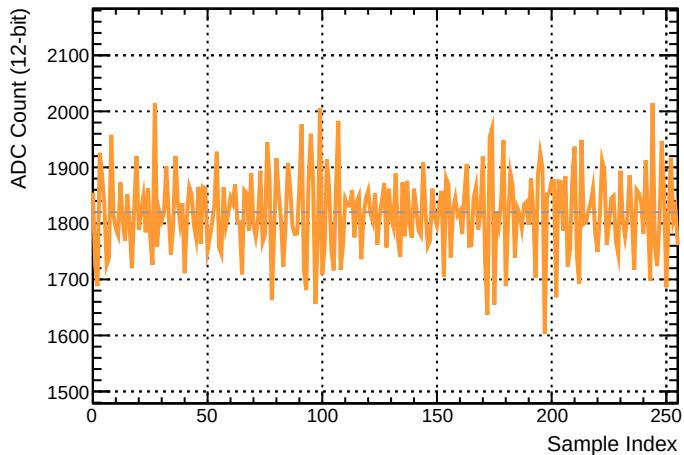
Ch 1 | Event #40 | Label: **SIGNAL** | CNN Score: 5474



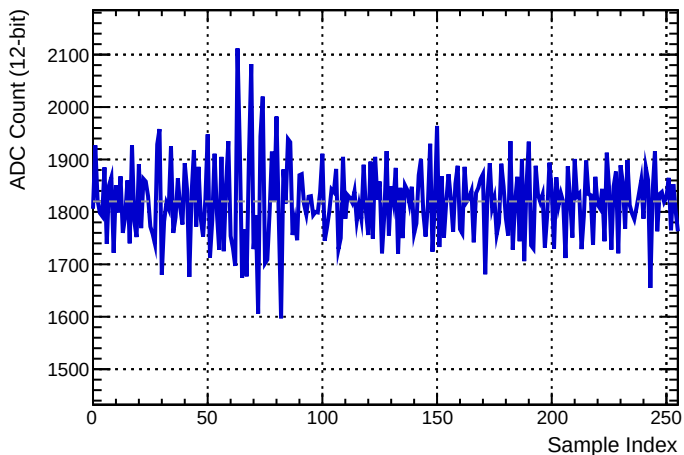
Ch 2 | Event #40 | Label: **SIGNAL** | CNN Score: 5474



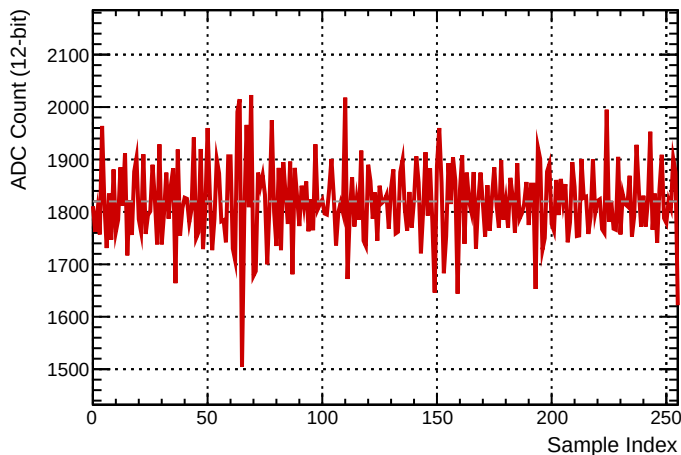
Ch 3 | Event #40 | Label: **SIGNAL** | CNN Score: 5474



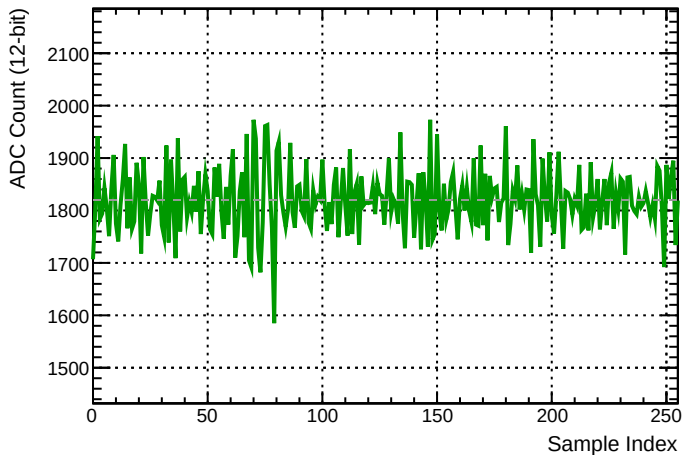
Ch 0 | Event #657 | Label: **SIGNAL** | CNN Score: 5616



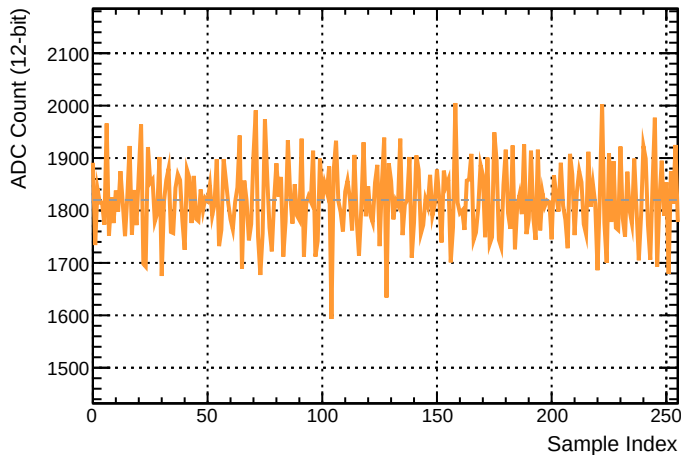
Ch 1 | Event #657 | Label: **SIGNAL** | CNN Score: 5616



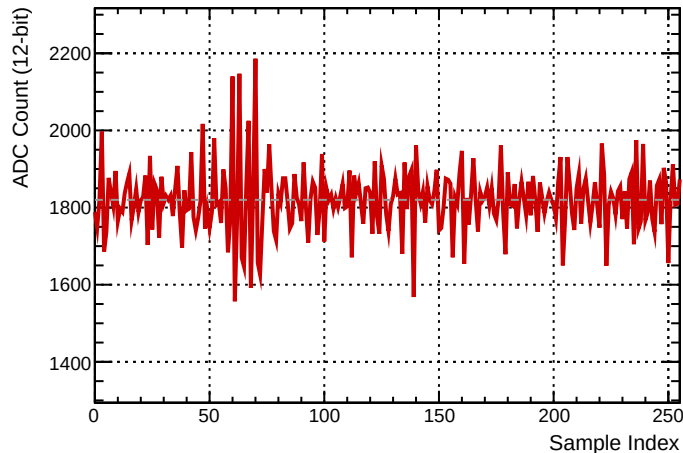
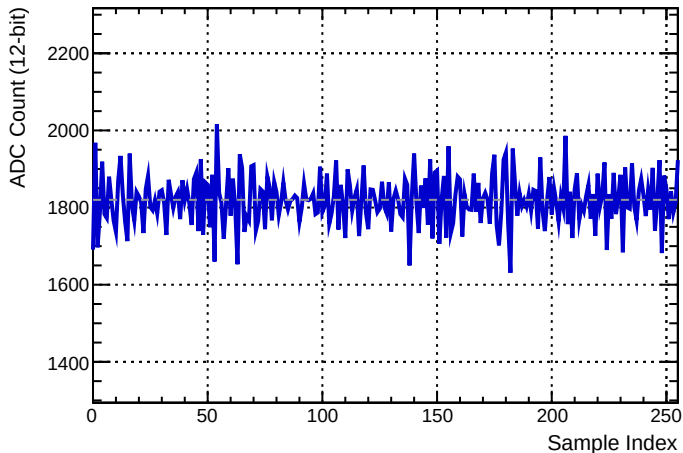
Ch 2 | Event #657 | Label: **SIGNAL** | CNN Score: 5616



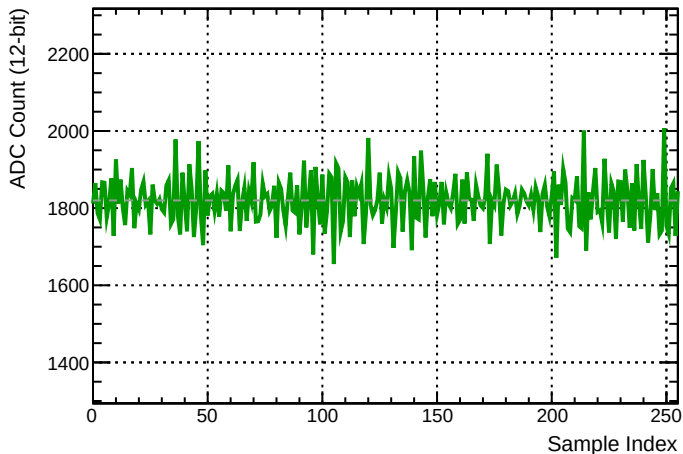
Ch 3 | Event #657 | Label: **SIGNAL** | CNN Score: 5616



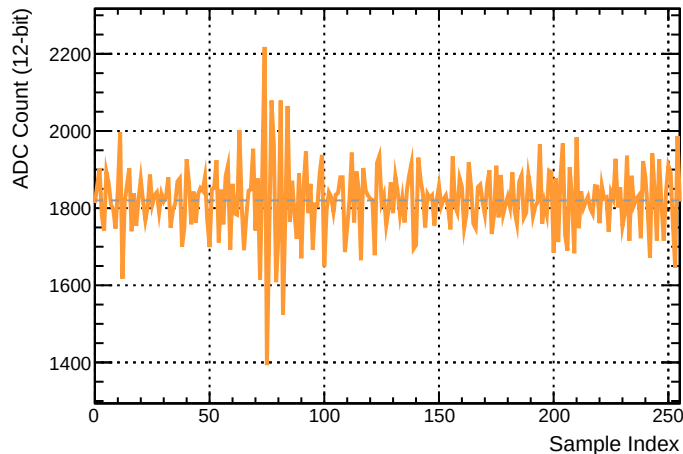
Ch 0 | Event #41 | Label: **SIGNAL** | CNN Score: 5481 | Dataset: Data

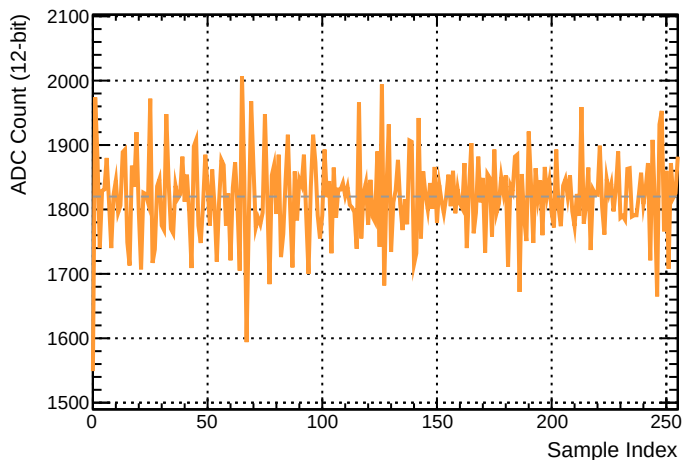
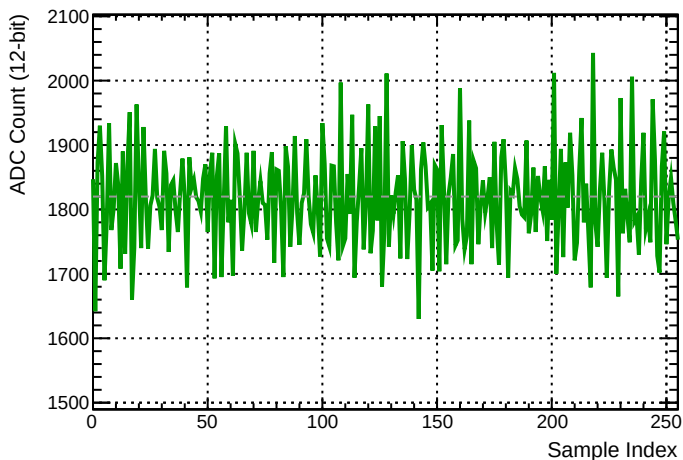
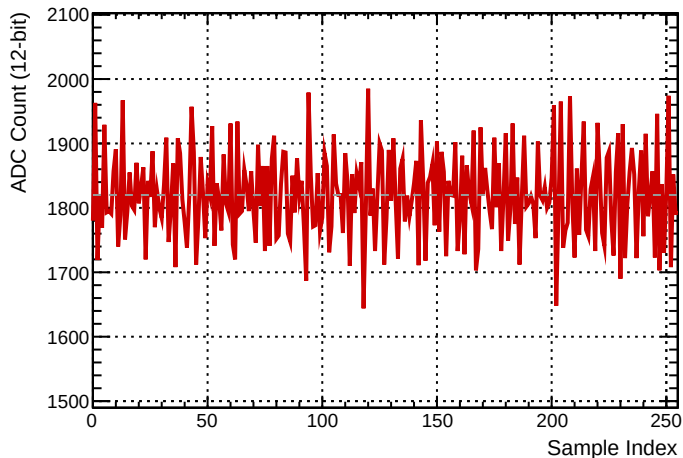
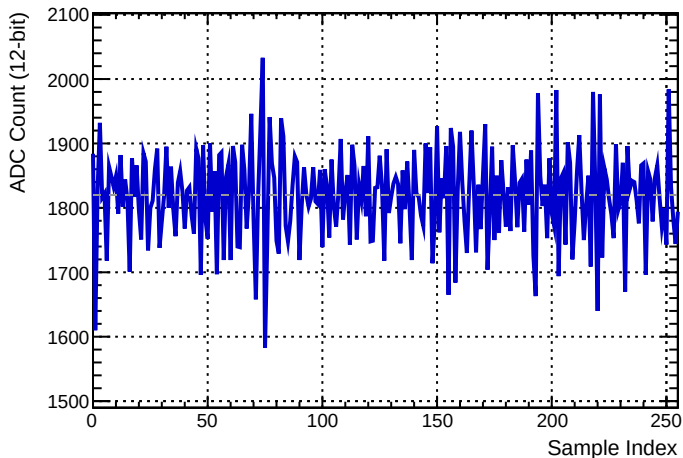


Ch 2 | Event #41 | Label: **SIGNAL** | CNN Score: 5481

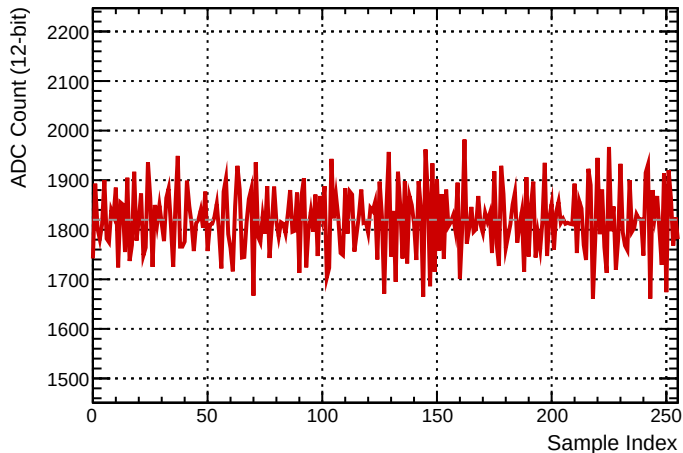
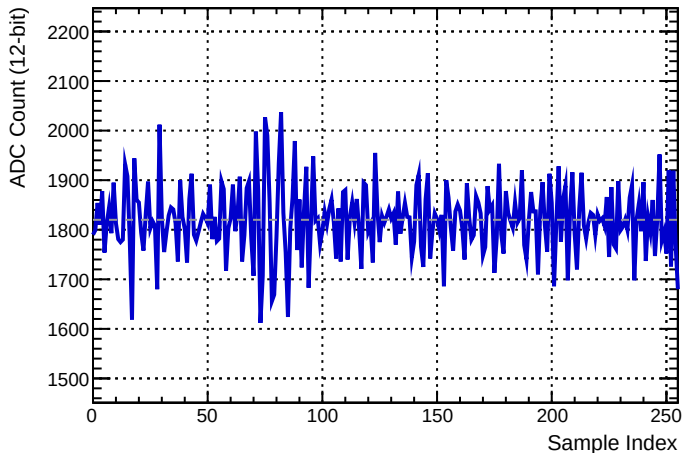


Ch 3 | Event #41 | Label: **SIGNAL** | CNN Score: 5481

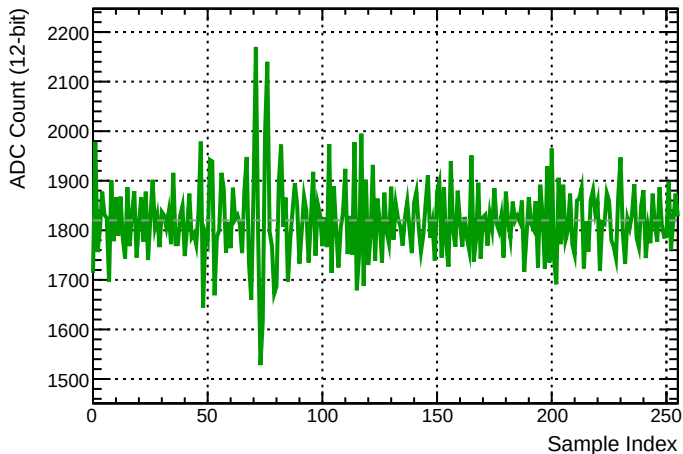




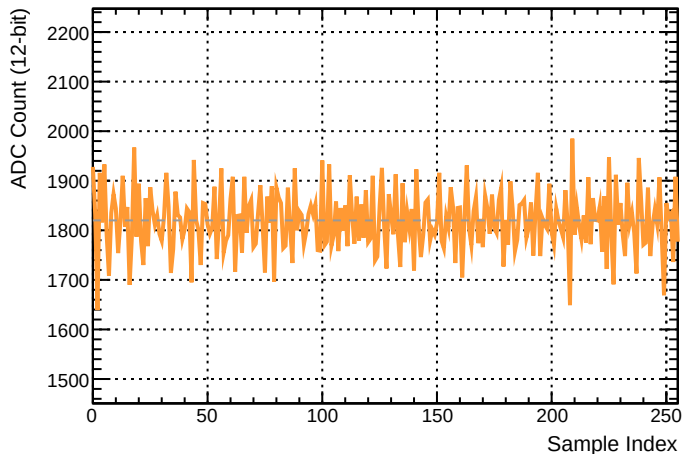
Ch 0 | Event #42 | Label: **SIGNAL** | CNN Score: 5251 | Dataset: 5251 | Event: 42 | Ch 0



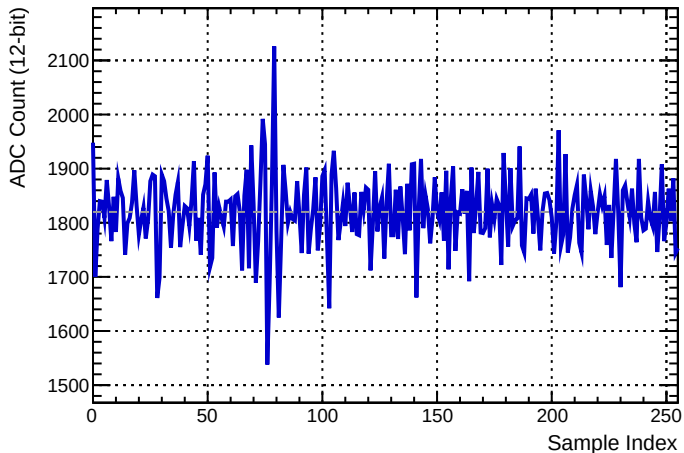
Ch 2 | Event #42 | Label: **SIGNAL** | CNN Score: 5251



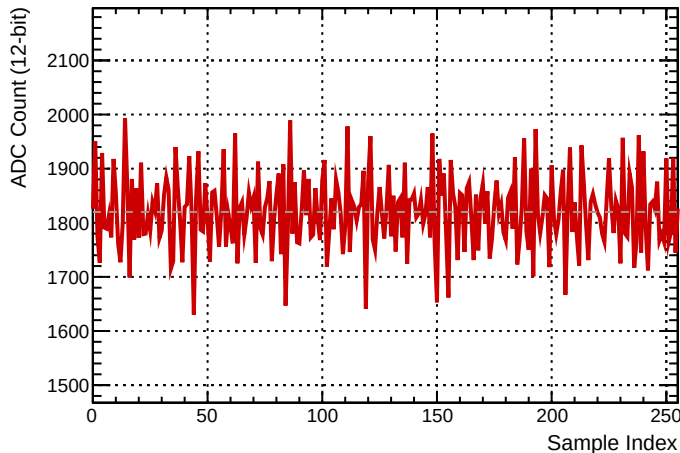
Ch 3 | Event #42 | Label: **SIGNAL** | CNN Score: 5251



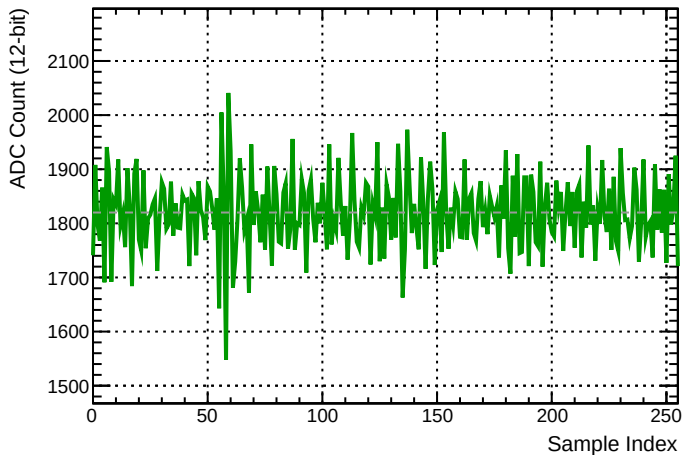
Ch 0 | Event #372 | Label: **SIGNAL** | CNN Score: 4929



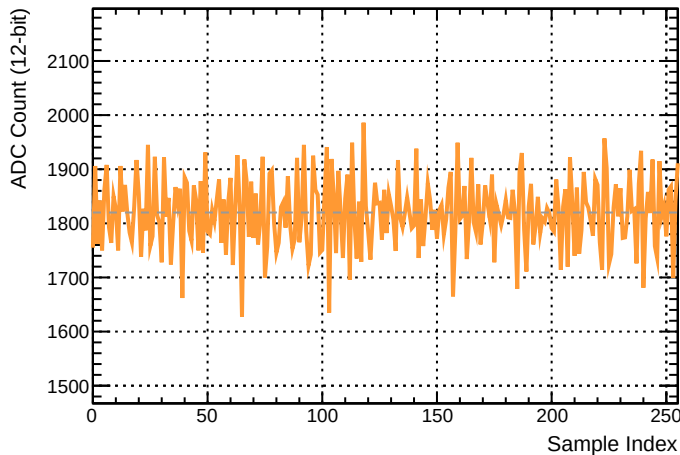
Ch 1 | Event #372 | Label: **SIGNAL** | CNN Score: 4929



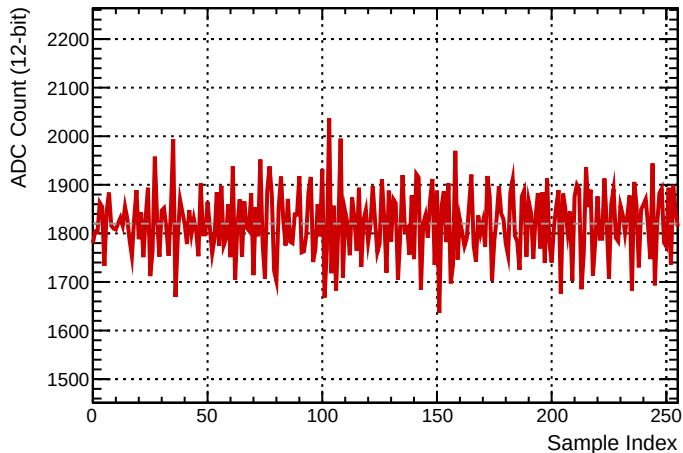
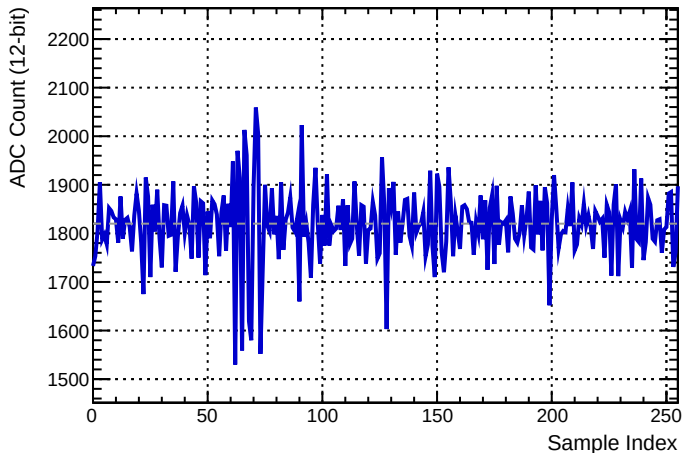
Ch 2 | Event #372 | Label: **SIGNAL** | CNN Score: 4929



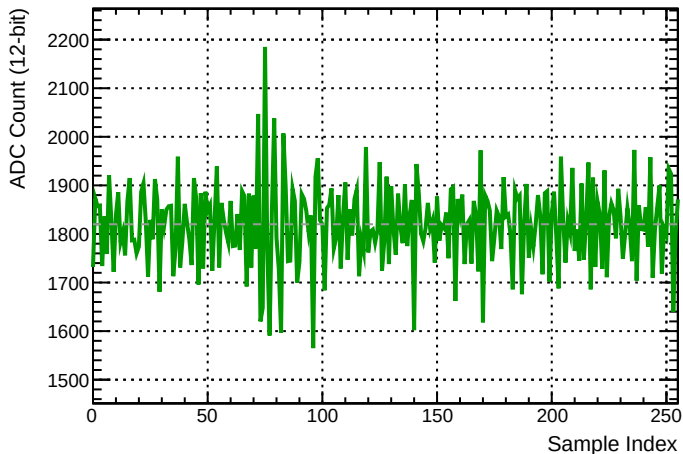
Ch 3 | Event #372 | Label: **SIGNAL** | CNN Score: 4929



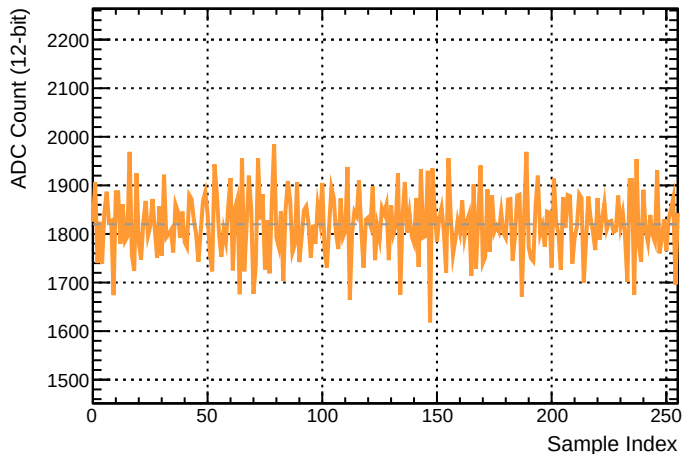
Ch 0 | Event #45 | Label: **SIGNAL** | CNN Score: 5191



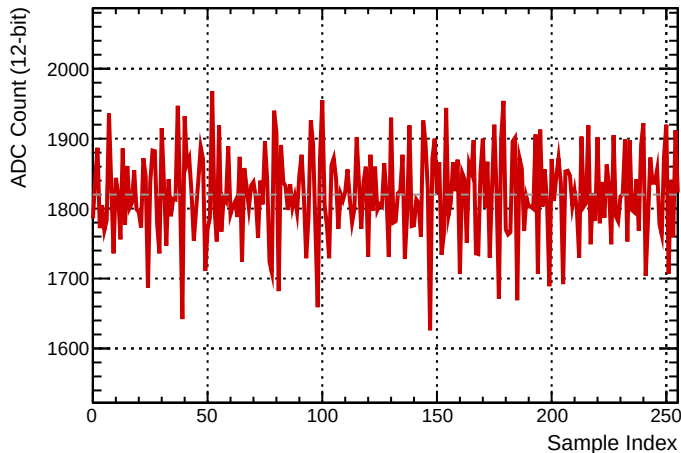
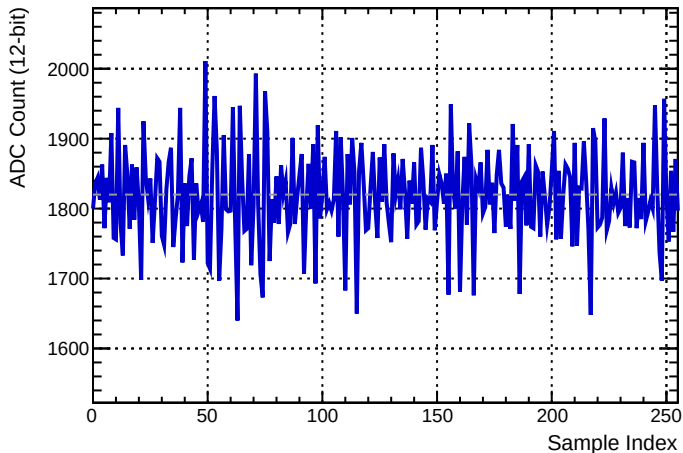
Ch 2 | Event #45 | Label: **SIGNAL** | CNN Score: 5191



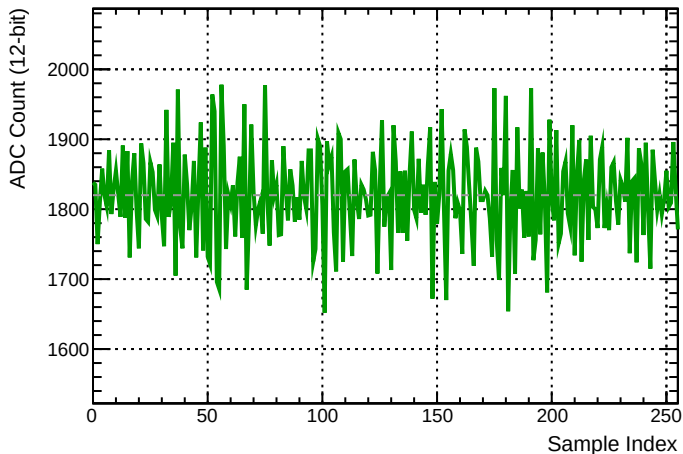
Ch 3 | Event #45 | Label: **SIGNAL** | CNN Score: 5191



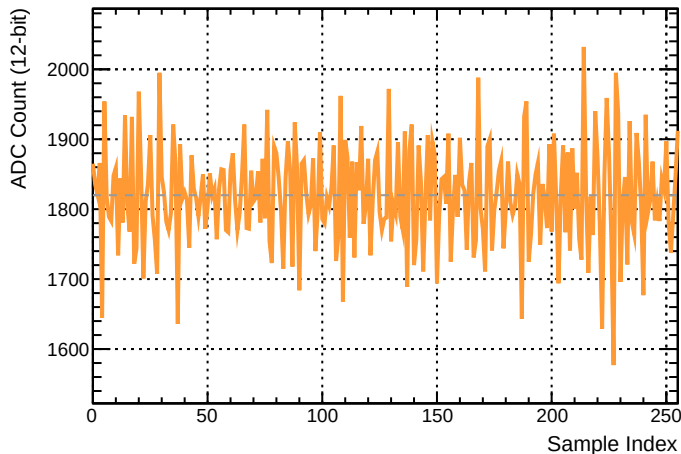
Ch 0 | Event #30 | Label: Background | Sim Test Set | Dataset Event #30 | Label: Background | CNN Score: 5351



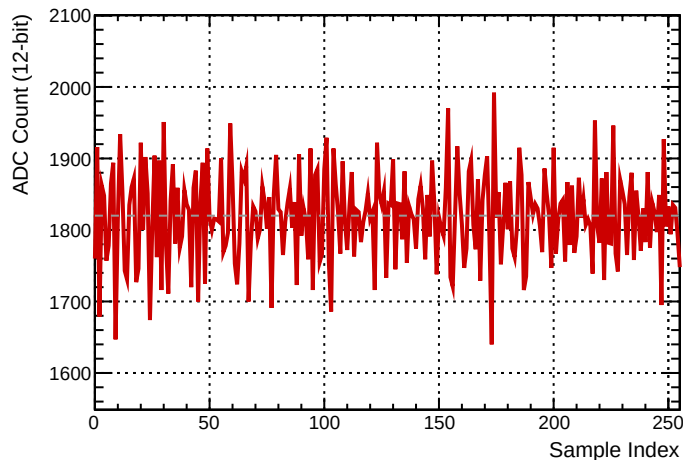
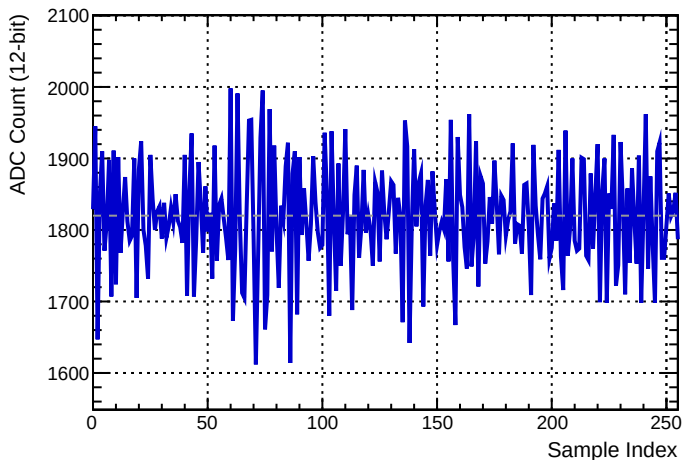
Ch 2 | Event #30 | Label: Background | CNN Score: 5351



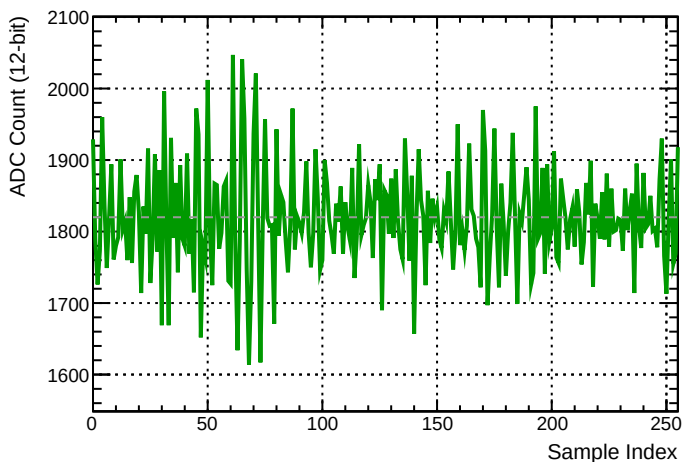
Ch 3 | Event #30 | Label: Background | CNN Score: 5351



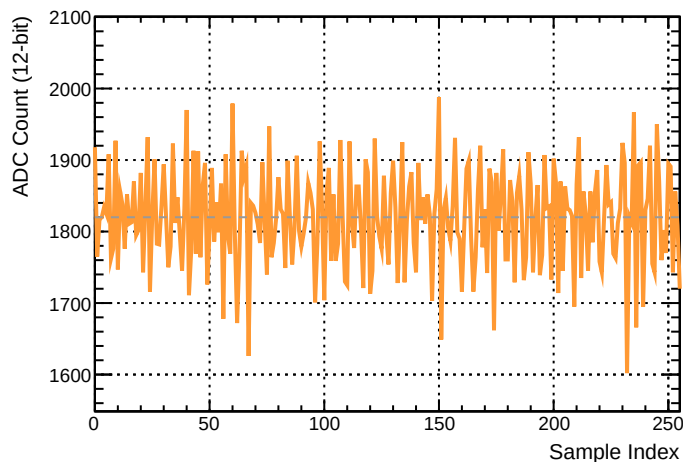
Ch 0 | Event #48 | Label: **SIGNAL** | CNN Score: 5065 | Dataset: **SIGNAL**

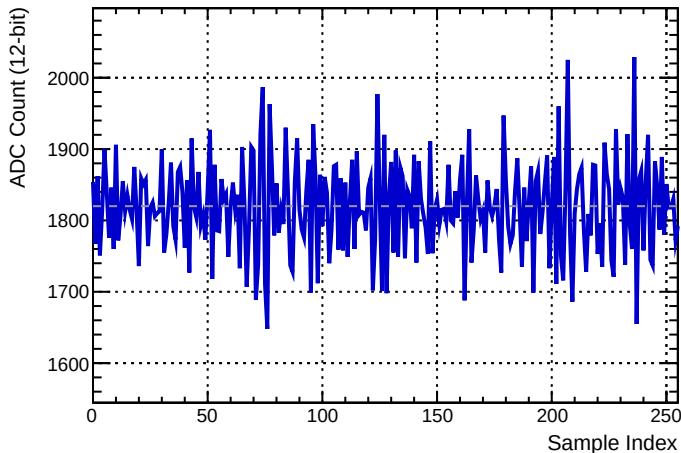
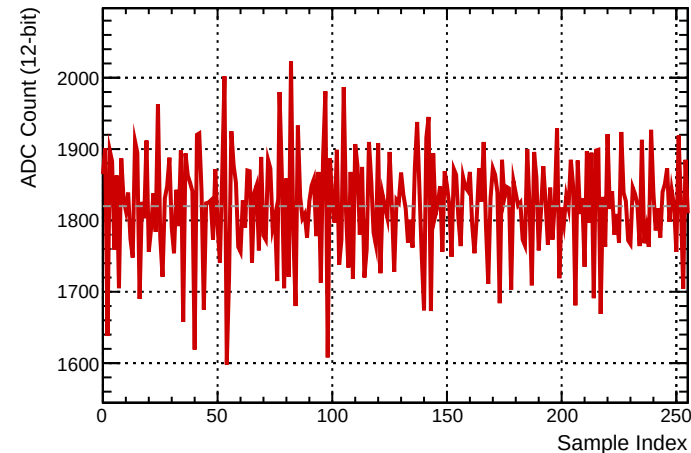
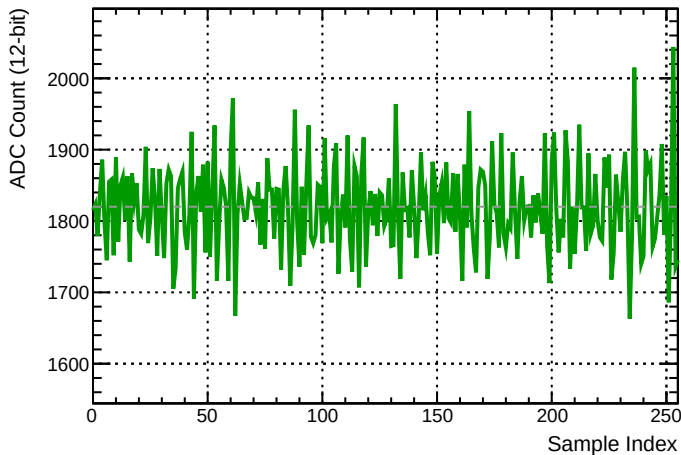
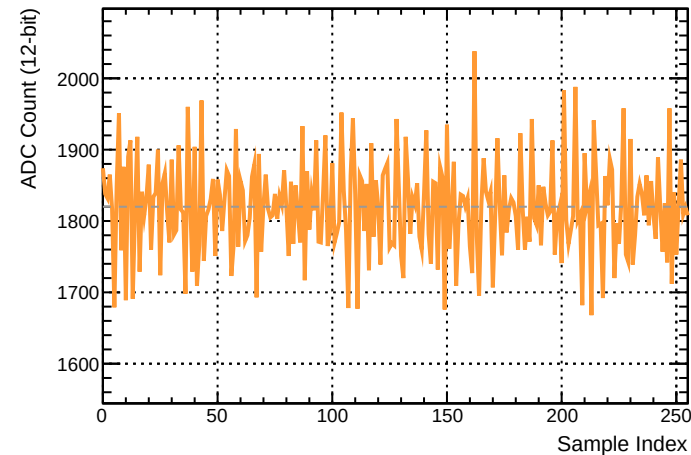


Ch 2 | Event #48 | Label: **SIGNAL** | CNN Score: 5065

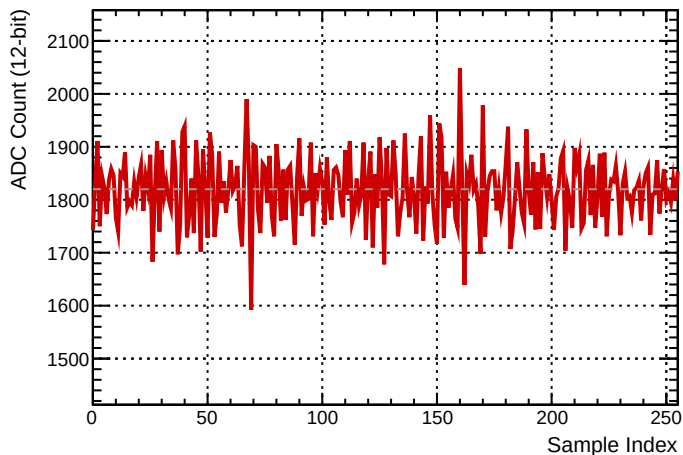
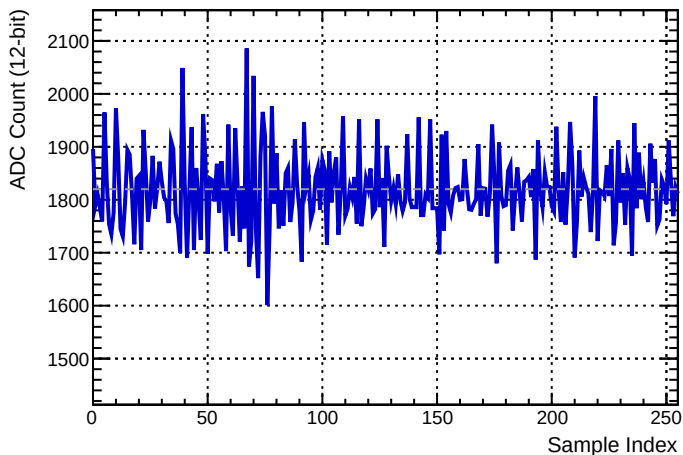


Ch 3 | Event #48 | Label: **SIGNAL** | CNN Score: 5065

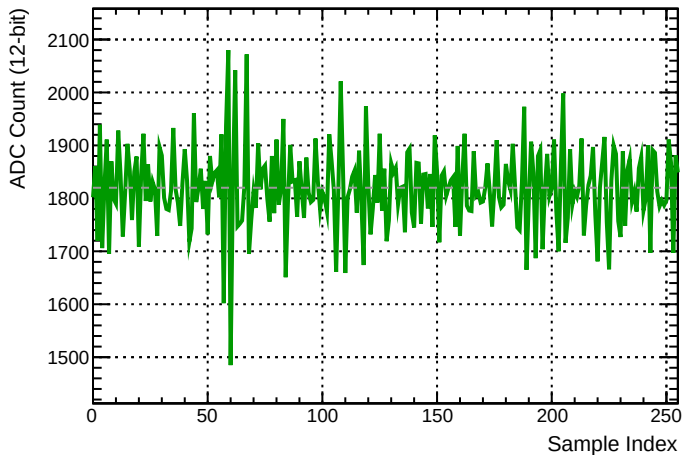


Ch 0 | Event #481 | Label: [Background](#) | CNN Score: 5103Ch 1 | Event #481 | Label: [Background](#) | CNN Score: 5103Ch 2 | Event #481 | Label: [Background](#) | CNN Score: 5103Ch 3 | Event #481 | Label: [Background](#) | CNN Score: 5103

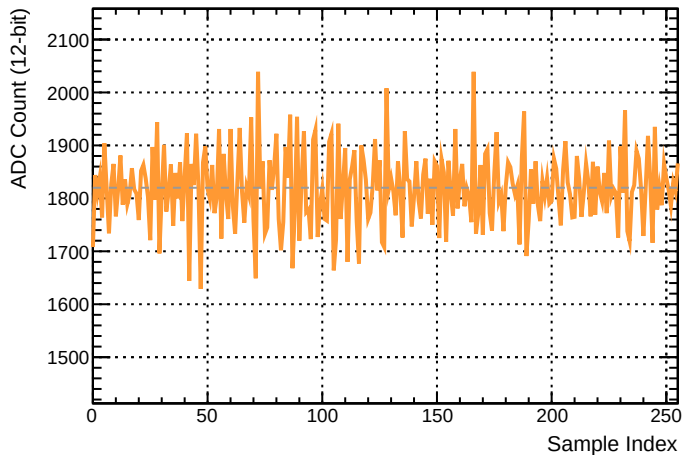
Ch 0 | Event #50 | Label: **SIGNAL** | CNN Score: 5139 | Dataset: Signal



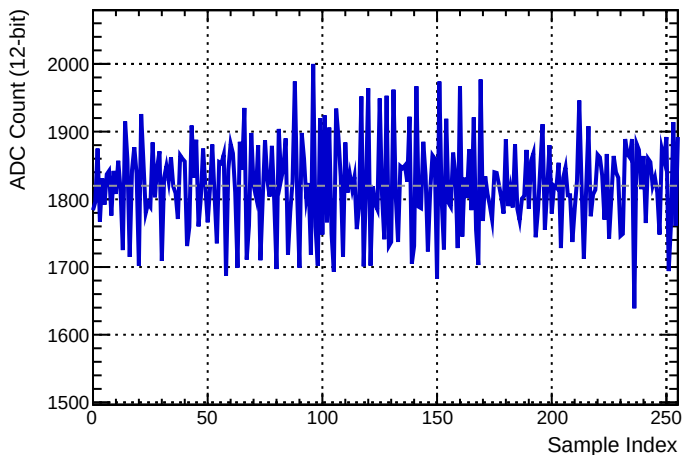
Ch 2 | Event #50 | Label: **SIGNAL** | CNN Score: 5139



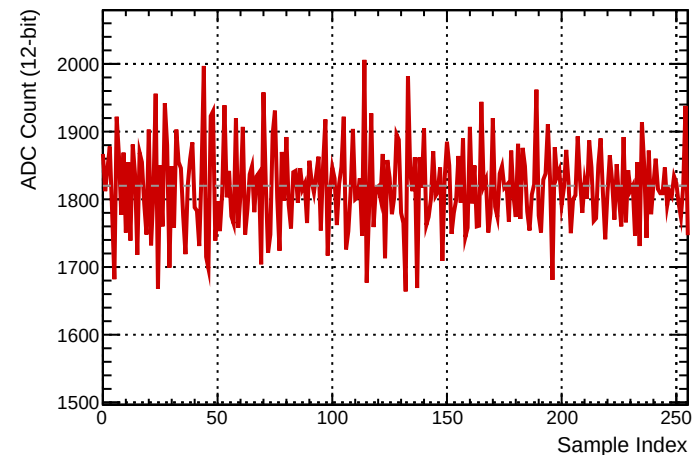
Ch 3 | Event #50 | Label: **SIGNAL** | CNN Score: 5139



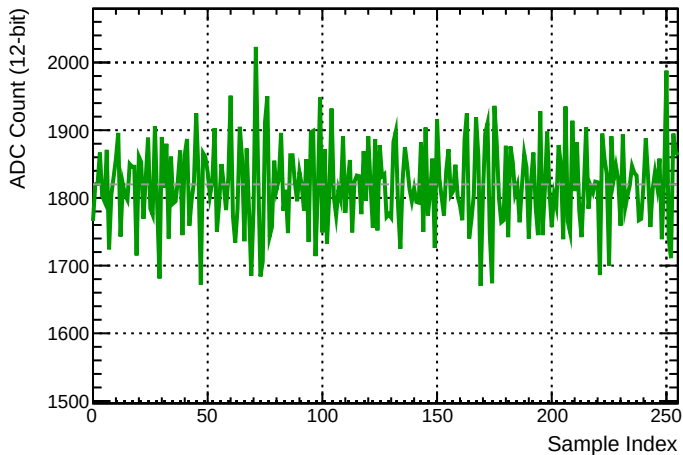
Ch 0 | Event #905 | Label: [Background](#) | CNN Score: 5515



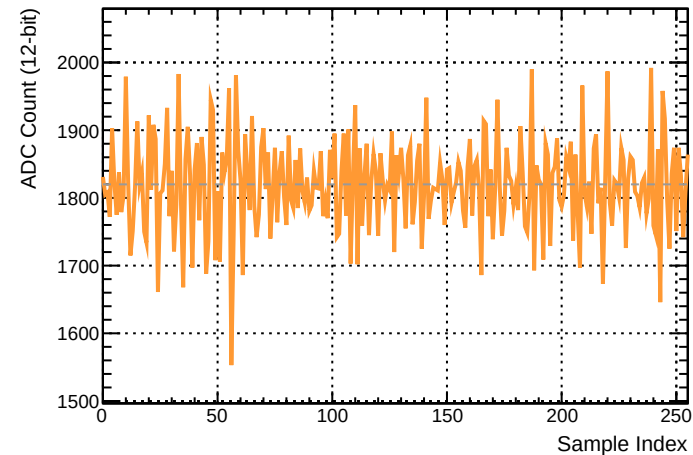
Ch 1 | Event #905 | Label: [Background](#) | CNN Score: 5515



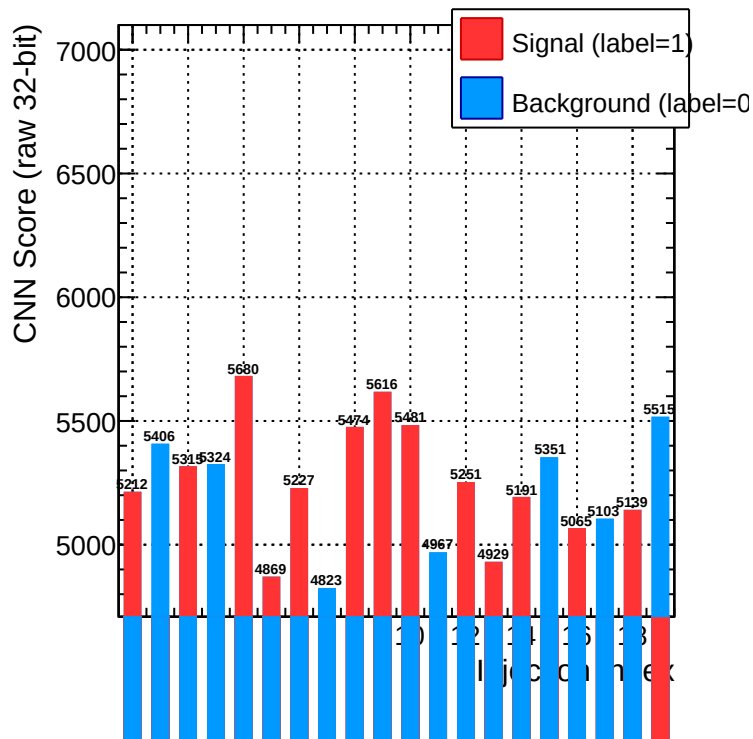
Ch 2 | Event #905 | Label: [Background](#) | CNN Score: 5515



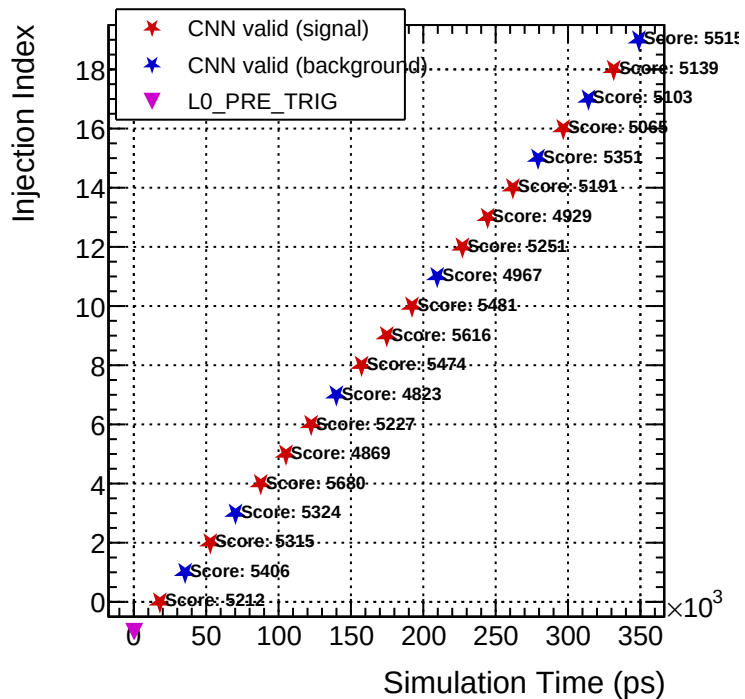
Ch 3 | Event #905 | Label: [Background](#) | CNN Score: 5515



CNN Output Score per Injected Event



Event Timeline



Simulation Results Summary (Page 1 / 2)

Test	EvIdx	Label	CNN Score	#Triggers	Result
0	#9	SIGNAL	5212 (0x145C)	0	Correct
1	#866	Background	5406 (0x151E)	0	Wrong
2	#17	SIGNAL	5315 (0x14C3)	0	Correct
3	#964	Background	5324 (0x14CC)	0	Wrong
4	#38	SIGNAL	5680 (0x1630)	0	Correct
5	#881	SIGNAL	4869 (0x1305)	0	Correct
6	#39	SIGNAL	5227 (0x146B)	0	Correct
7	#893	Background	4823 (0x12D7)	0	Wrong
8	#40	SIGNAL	5474 (0x1562)	0	Correct
9	#657	SIGNAL	5616 (0x15F0)	0	Correct

Simulation Results Summary (Page 2 / 2)

Test	EvIdx	Label	CNN Score	#Triggers	Result
10	#41	SIGNAL	5481 (0x1569)	0	Correct
11	#235	Background	4967 (0x1367)	0	Wrong
12	#42	SIGNAL	5251 (0x1483)	0	Correct
13	#372	SIGNAL	4929 (0x1341)	0	Correct
14	#45	SIGNAL	5191 (0x1447)	0	Correct
15	#30	Background	5351 (0x14E7)	0	Wrong
16	#48	SIGNAL	5065 (0x13C9)	0	Correct
17	#481	Background	5103 (0x13EF)	0	Wrong
18	#50	SIGNAL	5139 (0x1413)	0	Correct
19	#905	Background	5515 (0x158B)	0	Wrong

